

Introducing gEDA—The Open Source EDA Tool

gEDA offers a very mature suite of free software packages for complete project design cycle. However, there are three main roadblocks to the success of gEDA—use of Linux platform, discontinued gEDA manager development and documentation not being very recent. Here we discuss ways to overcome these hassles

ANKIT GUPTA

There are hundreds of good electronic design automation (EDA) tools available but most of them have to be purchased and the rest have limited-size design or limited functionalities. The only oasis in the desert is the world of Open Source, which is becoming more and more popular. Open Source EDA tools have long been considered as complicated, less user-friendly and limited in functionality, but it is time that you refresh your perception.

In this article I discuss gEDA project—an Open Source EDA that I recently evaluated and found worthy of being considered for use. gEDA offers a very mature suite of free software packages for complete project design cycle, which includes schematic capture, attribute editor, bill-of-materials (BOM) generation, netlist generation in various formats, simulation and PCB design. Though gEDA has some drawbacks too, it is a strong contender for becoming one of the most complete Open Source EDA.

There are three main reasons why designers are reluctant to use gEDA: Linux platform, discontinued gEDA manager development and documentation not being very recent. Here I suggest ways to address these hassles.

Linux platform

gEDA suite does not run on Windows platform currently and Windows support is not even a primary objective for gEDA. Windows is the most used

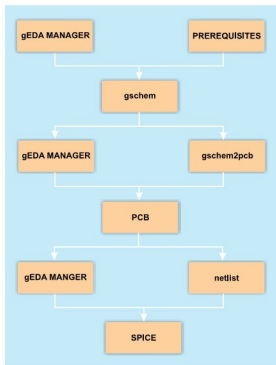


Fig. 1: gEDA workflow diagram

operating system around the world and switching to the Linux platform is not so quick. But, there is a solution to it—virtualisation software package. There are various free virtualisation software packages available, like VMware Player and Virtual Box, that can be used to run a guest operating system. Linux can easily be installed on this virtual environment without compromising Windows platform.

Once Linux is installed, gEDA packages can be installed on it. Installation procedure depends on the guest operating system used. The packages that need to be installed are gschem,

gschm2pcb, gattrib, pcb, gerbv, gnetlist and GTK-Wave. Installation can take quite some time and there is high chance that you overlook something during the installation if you are new to this environment. And if something does not work, you don't know whether it is because of you or the tool.

To avoid this hassle, download VMware ready-to-use image of Ubuntu. Installation of gEDA is very straightforward in Ubuntu. Open the downloaded virtual machine using VMware software (File→Open a Virtual Machine). Play the Virtual machine and you will have your Ubuntu running on the VMware software. Run Ubuntu Software center from the applications

menu. Search the packages one by one and install them.

The other way out could be a dual boot system. You can install Linux on the same system in a dual boot configuration. This means that you have to choose one operating system every time you reboot and cannot use both the operating systems at the same time like in virtualisation software. Again, you will have to install all the packages for gEDA suite in the Linux platform.

To avoid all the complexities related to package installation, you can alternatively choose to use FEL (Fedora Electronic Lab) prepared by EFY. FEL